

Notes on fraction graph constructions (v2)

1 An independent set in $E_{14/5}^{\boxtimes 3}$

Consider the set

$$S = \{t \cdot (1, 3, 9) : t \in \mathbb{Z}_{14}\} \subseteq \mathbb{Z}_{14}^3.$$

Explicitly, writing the elements as rows (with entries in $\mathbb{Z}_{14}$ that witness independence highlighted in green):

$$\begin{pmatrix} 0 & 0 & 0 \\ 1 & 3 & 9 \\ 2 & 6 & 4 \\ 3 & 9 & 13 \\ 4 & 12 & 8 \\ 5 & 1 & 3 \\ 6 & 4 & 12 \\ 7 & 7 & 7 \\ 8 & 10 & 2 \\ 9 & 13 & 11 \\ 10 & 2 & 6 \\ 11 & 5 & 1 \\ 12 & 8 & 10 \\ 13 & 11 & 5 \end{pmatrix}$$

One can verify that S is an independent set of size 14 in $E_{14/5}^{\boxtimes 3}$.

2 Baumert expansion to $E_{19/5}^{\boxtimes 3}$

We expand S to an independent set in $E_{19/5}^{\boxtimes 3}$ as follows. Note that $19 = 14 + 5$. Given an element $(a_1, a_2, a_3) \in S \subseteq \mathbb{Z}_{14}^3$, we map each coordinate into $\mathbb{Z}_{19}$ by the rule:

- If $a_i \in \{0, 1, 2, 3, 4\}$: replace a_i by *both* a_i and $a_i + 5$ in $\mathbb{Z}_{19}$.
- If $a_i \in \{5, 6, \dots, 13\}$: replace a_i by the single value $a_i + 5$ in $\mathbb{Z}_{19}$.

Each element of S thus produces 2^k elements in $\mathbb{Z}_{19}^3$, where k is the number of coordinates in $\{0, 1, 2, 3, 4\}$. For example, $(0, 0, 0)$ produces $2^3 = 8$ elements (all of $\{0, 5\}^3$), while $(7, 7, 7)$ produces just one element $(12, 12, 12)$.

Applying this to all of S yields the following independent set S' of size 36 in $E_{19/5}^{\boxtimes 3}$

(grouped by parent element in S):

$$\begin{aligned}
(0, 0, 0) &\rightsquigarrow (0, 0, 0), (0, 0, 5), (0, 5, 0), (0, 5, 5), (5, 0, 0), (5, 0, 5), (5, 5, 0), (5, 5, 5) \\
(1, 3, 9) &\rightsquigarrow (1, 3, 14), (1, 8, 14), (6, 3, 14), (6, 8, 14) \\
(2, 6, 4) &\rightsquigarrow (2, 11, 4), (2, 11, 9), (7, 11, 4), (7, 11, 9) \\
(3, 9, 13) &\rightsquigarrow (3, 14, 18), (8, 14, 18) \\
(4, 12, 8) &\rightsquigarrow (4, 17, 13), (9, 17, 13) \\
(5, 1, 3) &\rightsquigarrow (10, 1, 3), (10, 1, 8), (10, 6, 3), (10, 6, 8) \\
(6, 4, 12) &\rightsquigarrow (11, 4, 17), (11, 9, 17) \\
(7, 7, 7) &\rightsquigarrow (12, 12, 12) \\
(8, 10, 2) &\rightsquigarrow (13, 15, 2), (13, 15, 7) \\
(9, 13, 11) &\rightsquigarrow (14, 18, 16) \\
(10, 2, 6) &\rightsquigarrow (15, 2, 11), (15, 7, 11) \\
(11, 5, 1) &\rightsquigarrow (16, 10, 1), (16, 10, 6) \\
(12, 8, 10) &\rightsquigarrow (17, 13, 15) \\
(13, 11, 5) &\rightsquigarrow (18, 16, 10)
\end{aligned}$$

This is the *Baumert expansion*. One can verify that S' is independent in $E_{19/5}^{\boxtimes 3}$.

3 Coordinate-wise reduction

3.1 Perturbation

Fix a coordinate i and define a directed graph G_i on the vertex set S' with a directed edge $x \rightarrow y$ whenever:

1. $x_i + q \equiv y_i \pmod{p}$ (the i -th coordinate shifts by q), and
2. $\text{dist}(x_j, y_j) < q$ for all $j \neq i$ (the other coordinates are *strictly* close),

where p and q are the modulus and adjacency threshold for the respective coordinates. If G_i is acyclic, define $n_{i,k}$ as the length of the longest directed path ending at vertex k , and $M_i = \max_k n_{i,k}$. We perturb coordinate i alone:

$$x_i \mapsto (1+M_i)x_i + n_{i,x} \pmod{p(1+M_i)},$$

leaving all other coordinates unchanged. Coordinate i moves from $E_{p/q}$ to $E_{p(1+M_i)/(q(1+M_i)+1)}$.

We apply this to coordinate 1 of $S' \subseteq E_{19/5}^{\boxtimes 3}$. The graph G_1 has 40 directed edges, is acyclic, and has $M_1 = 9$ (see Figure 1).

The perturbation gives:

$$x_1 \mapsto 10x_1 + n_{1,x} \pmod{190}.$$

Coordinate 1 moves from $E_{19/5}$ to $E_{190/51}$, while coordinates 2, 3 remain in $E_{19/5}$. The independent set now lives in $E_{190/51} \boxtimes E_{19/5} \boxtimes E_{19/5}$:

1: (5, 0, 0)	10: (14, 8, 14)	19: (42, 17, 13)	28: (137, 15, 2)
2: (0, 0, 5)	11: (65, 3, 14)	20: (93, 17, 13)	29: (130, 15, 7)
3: (0, 5, 0)	12: (65, 8, 14)	21: (107, 1, 3)	30: (144, 18, 16)
4: (0, 5, 5)	13: (28, 11, 4)	22: (102, 1, 8)	31: (153, 2, 11)
5: (56, 0, 0)	14: (21, 11, 9)	23: (102, 6, 3)	32: (153, 7, 11)
6: (51, 0, 5)	15: (79, 11, 4)	24: (102, 6, 8)	33: (167, 10, 1)
7: (51, 5, 0)	16: (72, 11, 9)	25: (116, 4, 17)	34: (160, 10, 6)
8: (51, 5, 5)	17: (35, 14, 18)	26: (116, 9, 17)	35: (174, 13, 15)
9: (14, 3, 14)	18: (86, 14, 18)	27: (123, 12, 12)	36: (181, 16, 10)

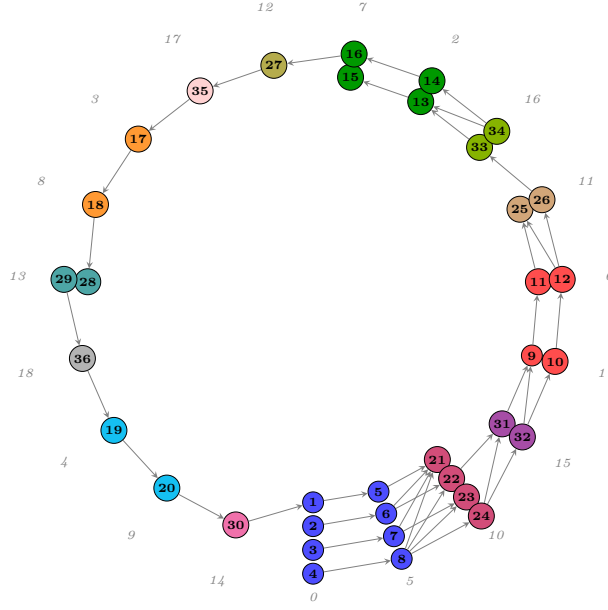


Figure 1: The strict shift graph G_1 for $S' \subseteq E_{19/5}^{\boxtimes 3}$ (40 edges, acyclic, $M_1 = 9$). Vertices arranged by residue class of coordinate 1 along the $+5$ cycle.

3.2 Compression

After perturbation, not all residues in the new modulus are used. To compress, list the distinct values of coordinate i in cyclic order and record for each one how many of the *next* values it is adjacent to (the *forward adjacency count*). Whenever the count drops from c_k to $c_{k+1} = c_k - 1$, the neighbourhood of v_{k+1} is contained in $N(v_k) \cup \{v_k\}$, so the map $v_k \mapsto v_{k+1}$ is a valid cohomomorphism. Remove v_k and update $c_{k-1} \rightarrow c_{k-1} - 1$. Repeat until the forward adjacency sequence is uniform with value $q' - 1$; the coordinate is now $E_{m/q'}$ where m is the number of surviving values.

For coordinate 1 after perturbation to $E_{190/51}$, the 36 elements use 26 distinct values in coordinate 1 with forward adjacency sequence uniformly equal to 6. This means $E_{190/51}[S_1] \cong E_{26/7}$.

The independent set now lives in $E_{26/7} \boxtimes E_{19/5} \boxtimes E_{19/5}$, where $26/7 \approx 3.714 < 19/5 = 3.8$. Explicitly (coordinate 1 in $\mathbb{Z}_{26}$, coordinates 2, 3 in $\mathbb{Z}_{19}$):

1: (1, 0, 0)	10: (2, 8, 14)	19: (6, 17, 13)	28: (19, 15, 2)
2: (0, 0, 5)	11: (9, 3, 14)	20: (13, 17, 13)	29: (18, 15, 7)
3: (0, 5, 0)	12: (9, 8, 14)	21: (15, 1, 3)	30: (20, 18, 16)
4: (0, 5, 5)	13: (4, 11, 4)	22: (14, 1, 8)	31: (21, 2, 11)
5: (8, 0, 0)	14: (3, 11, 9)	23: (14, 6, 3)	32: (21, 7, 11)
6: (7, 0, 5)	15: (11, 11, 4)	24: (14, 6, 8)	33: (23, 10, 1)
7: (7, 5, 0)	16: (10, 11, 9)	25: (16, 4, 17)	34: (22, 10, 6)
8: (7, 5, 5)	17: (5, 14, 18)	26: (16, 9, 17)	35: (24, 13, 15)
9: (2, 3, 14)	18: (12, 14, 18)	27: (17, 12, 12)	36: (25, 16, 10)

4 Second reduction of coordinate 1

We apply the process again to coordinate 1, now in $E_{26/7}$. The graph G_1 (shift by $+7 \bmod 26$, strict distance < 5 in coordinates 2, 3) has 38 edges, is acyclic, with $M_1 = 17$. The

perturbation maps coordinate 1 from $E_{26/7}$ to $E_{468/127}$. The 36 elements use 33 distinct values with forward adjacency sequence

8, 9, 8, 8, 7, 7, 8, 8, 8, 8, 8, 8, 8, 8.

After 11 removal steps, this reduces to 22 values with uniform count 5, giving $E_{468/127}[S_1] \cong E_{22/6}$. The independent set now lives in $E_{22/6} \boxtimes E_{19/5} \boxtimes E_{19/5}$:

1: (2, 0, 0)	10: (2, 8, 14)	19: (6, 17, 13)	28: (17, 15, 2)
2: (1, 0, 5)	11: (9, 3, 14)	20: (12, 17, 13)	29: (16, 15, 7)
3: (0, 5, 0)	12: (8, 8, 14)	21: (14, 1, 3)	30: (18, 18, 16)
4: (0, 5, 5)	13: (4, 11, 4)	22: (13, 1, 8)	31: (19, 2, 11)
5: (8, 0, 0)	14: (3, 11, 9)	23: (12, 6, 3)	32: (18, 7, 11)
6: (7, 0, 5)	15: (10, 11, 4)	24: (12, 6, 8)	33: (20, 10, 1)
7: (6, 5, 0)	16: (9, 11, 9)	25: (15, 4, 17)	34: (19, 10, 6)
8: (6, 5, 5)	17: (5, 14, 18)	26: (14, 9, 17)	35: (21, 13, 15)
9: (3, 3, 14)	18: (11, 14, 18)	27: (15, 12, 12)	36: (0, 16, 10)

5 Further reductions of coordinate 1

Applying the process once more to coordinate 1, now in $E_{22/6}$: the graph G_1 (shift by +6 mod 22, strict distance <5 in coordinates 2, 3) has 34 edges, is acyclic, with $M_1 = 17$. The perturbation maps coordinate 1 from $E_{22/6}$ to $E_{396/109}$. The 33 distinct values have forward adjacency sequence

8, 9, 9, 9, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8.

After 4 removal steps, this reduces to 29 values with uniform count 7, giving $E_{396/109}[S_1] \cong E_{29/8}$. The independent set in $E_{29/8} \boxtimes E_{19/5} \boxtimes E_{19/5}$ is:

1: (4, 0, 0)	10: (3, 8, 14)	19: (9, 17, 13)	28: (23, 15, 2)
2: (2, 0, 5)	11: (13, 3, 14)	20: (17, 17, 13)	29: (22, 15, 7)
3: (0, 5, 0)	12: (11, 8, 14)	21: (20, 1, 3)	30: (25, 18, 16)
4: (0, 5, 5)	13: (6, 11, 4)	22: (18, 1, 8)	31: (26, 2, 11)
5: (12, 0, 0)	14: (4, 11, 9)	23: (16, 6, 3)	32: (24, 7, 11)
6: (10, 0, 5)	15: (14, 11, 4)	24: (16, 6, 8)	33: (27, 10, 1)
7: (8, 5, 0)	16: (12, 11, 9)	25: (21, 4, 17)	34: (25, 10, 6)
8: (8, 5, 5)	17: (7, 14, 18)	26: (19, 9, 17)	35: (28, 13, 15)
9: (5, 3, 14)	18: (15, 14, 18)	27: (20, 12, 12)	36: (1, 16, 10)

Applying once more: G_1 on $E_{29/8}$ has 35 edges, is acyclic, with $M_1 = 17$. The perturbation maps coordinate 1 to $E_{522/145}$. All 36 values are distinct and the forward adjacency sequence is already uniform at 9, so $E_{522/145}[S_1] \cong E_{36/10}$. The independent set in $E_{36/10} \boxtimes E_{19/5} \boxtimes E_{19/5}$ is:

1: (6, 0, 0)	10: (4, 8, 14)	19: (12, 17, 13)	28: (29, 15, 2)
2: (3, 0, 5)	11: (17, 3, 14)	20: (22, 17, 13)	29: (28, 15, 7)
3: (1, 5, 0)	12: (14, 8, 14)	21: (26, 1, 3)	30: (32, 18, 16)
4: (0, 5, 5)	13: (8, 11, 4)	22: (23, 1, 8)	31: (33, 2, 11)
5: (16, 0, 0)	14: (5, 11, 9)	23: (21, 6, 3)	32: (30, 7, 11)
6: (13, 0, 5)	15: (18, 11, 4)	24: (20, 6, 8)	33: (34, 10, 1)
7: (11, 5, 0)	16: (15, 11, 9)	25: (27, 4, 17)	34: (31, 10, 6)
8: (10, 5, 5)	17: (9, 14, 18)	26: (24, 9, 17)	35: (35, 13, 15)
9: (7, 3, 14)	18: (19, 14, 18)	27: (25, 12, 12)	36: (2, 16, 10)

At this point, G_1 on $E_{36/10}$ has 36 directed edges and contains directed cycles. The process terminates for coordinate 1.

In summary, the independent set of size 36 has moved through:

$$\begin{aligned}
E_{14/5}^{\boxtimes 3} &\xrightarrow{\text{Baumert}} E_{19/5}^{\boxtimes 3} \\
&\xrightarrow{\text{perturb+compress}} E_{26/7} \boxtimes E_{19/5} \boxtimes E_{19/5} \\
&\xrightarrow{\text{perturb+compress}} E_{22/6} \boxtimes E_{19/5} \boxtimes E_{19/5} \\
&\xrightarrow{\text{perturb+compress}} E_{29/8} \boxtimes E_{19/5} \boxtimes E_{19/5} \\
&\xrightarrow{\text{perturb+compress}} E_{36/10} \boxtimes E_{19/5} \boxtimes E_{19/5}.
\end{aligned}$$

The fractions in coordinate 1 are $19/5 = 3.80$, $26/7 \approx 3.71$, $22/6 \approx 3.67$, $29/8 = 3.625$, $36/10 = 3.60$.

Applying the same process to coordinates 2 and 3 in turn, each follows the identical reduction sequence $E_{19/5} \rightarrow E_{26/7} \rightarrow E_{22/6} \rightarrow E_{29/8} \rightarrow E_{36/10}$, at which point G_i becomes cyclic. The final independent set of size 36 lives in $E_{36/10}^{\boxtimes 3} = E_{18/5}^{\boxtimes 3}$:

1: (6, 3, 0)	10: (4, 18, 29)	19: (12, 34, 25)	28: (29, 33, 7)
2: (3, 0, 10)	11: (17, 9, 31)	20: (22, 35, 28)	29: (28, 30, 17)
3: (1, 13, 3)	12: (14, 19, 32)	21: (26, 7, 8)	30: (32, 2, 33)
4: (0, 10, 13)	13: (8, 26, 9)	22: (23, 4, 18)	31: (33, 5, 23)
5: (16, 6, 5)	14: (5, 21, 19)	23: (21, 17, 11)	32: (30, 15, 24)
6: (13, 1, 15)	15: (18, 27, 12)	24: (20, 14, 21)	33: (34, 23, 4)
7: (11, 16, 6)	16: (15, 24, 22)	25: (27, 12, 34)	34: (31, 20, 14)
8: (10, 11, 16)	17: (9, 29, 35)	26: (24, 22, 1)	35: (35, 28, 30)
9: (7, 8, 26)	18: (19, 32, 2)	27: (25, 25, 27)	36: (2, 31, 20)